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Ex. No.: 1

CAPTURE FLAGS-ENCRYPTION CRYPTO 101

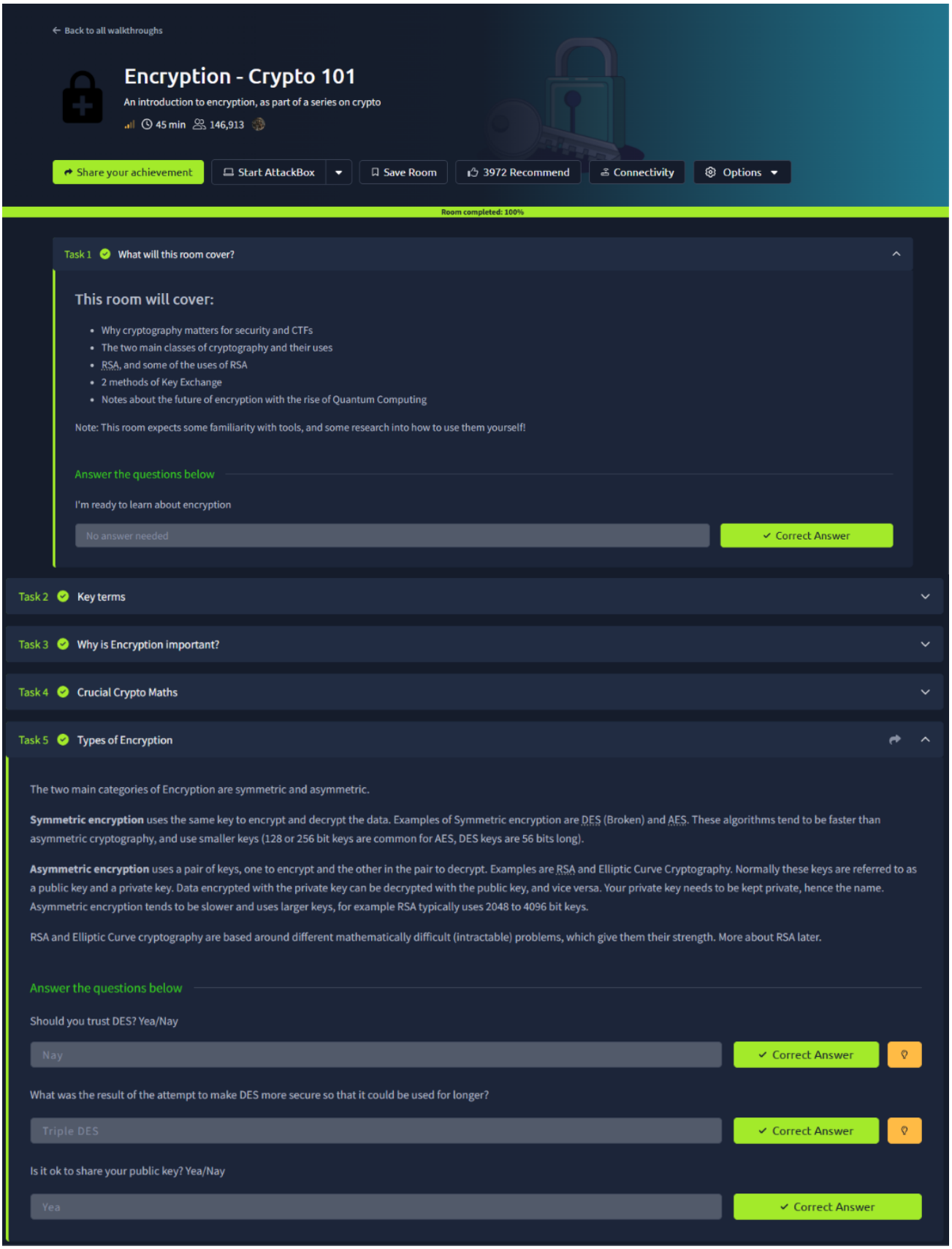
Aim:

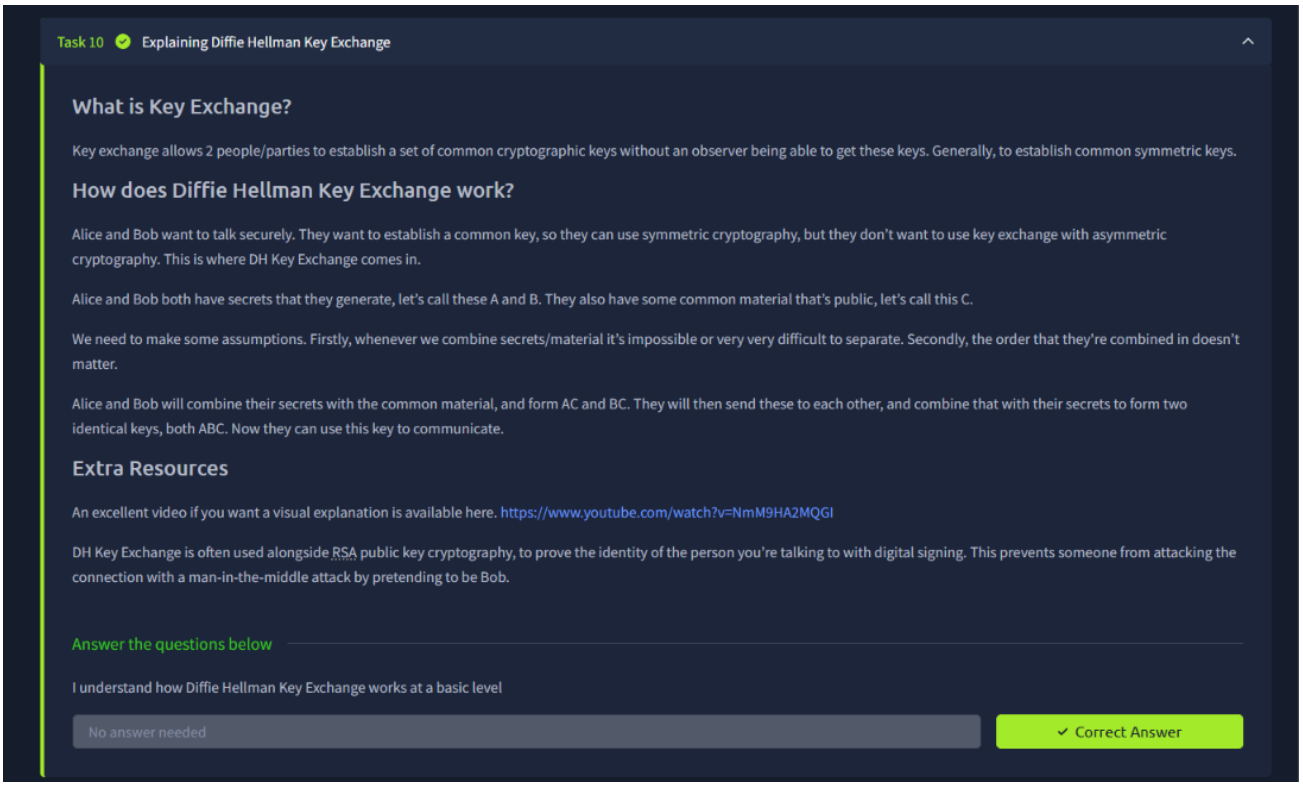
To capture the various flags in Encryption Crypto101 in TryHackMe platform.

Algorithm:

1. Open the Passive Reconnaissance lab on TryHackMe using the provided link.
2. Launch the AttackBox to start the Kali Linux environment.
3. Solve the mathematical calculations involved in RSA cryptography.
4. Identify the Certificate Authority (CA) that issued the HTTPS certificate for tryhackme.com.
5. Generate an SSH public and private key pair using ssh-keygen and use it for SSH authentication.
6. Decrypt the provided GPG-encrypted file and determine the hidden secret word.

Output:


← Back to all walkthroughs
Encryption - Crypto 101
An introduction to encryption, as part of a series on crypto
45 min
146,913
Share your achievement
Start AttackBox
Save Room
3972 Recommend
Connectivity
Options
Room completed: 100%
Task 1 What will this room cover?
This room will cover:
Why cryptography matters for security and CTFs
The two main classes of cryptography and their uses
RSA, and some of the uses of RSA
2 methods of Key Exchange
Notes about the future of encryption with the rise of Quantum Computing
Note: This room expects some familiarity with tools, and some research into how to use them yourself!
Answer the questions below
I'm ready to learn about encryption
No answer needed
Correct Answer
Task 2 Key terms
Task 3 Why is Encryption important?
Task 4 Crucial Crypto Maths
Task 5 Types of Encryption
The two main categories of Encryption are symmetric and asymmetric.
Symmetric encryption uses the same key to encrypt and decrypt the data. Examples of Symmetric encryption are DES (Broken) and AES. These algorithms tend to be faster than asymmetric cryptography, and use smaller keys (128 or 256 bit keys are common for AES, DES keys are 56 bits long).
Asymmetric encryption uses a pair of keys, one to encrypt and the other in the pair to decrypt. Examples are RSA and Elliptic Curve Cryptography. Normally these keys are referred to as a public key and a private key. Data encrypted with the private key can be decrypted with the public key, and vice versa. Your private key needs to be kept private, hence the name. Asymmetric encryption tends to be slower and uses larger keys, for example RSA typically uses 2048 to 4096 bit keys.
RSA and Elliptic Curve cryptography are based around different mathematically difficult (intractable) problems, which give them their strength. More about RSA later.
Answer the questions below
Should you trust DES? Yea/Nay
Nay
Correct Answer
What was the result of the attempt to make DES more secure so that it could be used for longer?
Triple DES
Correct Answer
Is it ok to share your public key? Yea/Nay
Yea
Correct Answer


Task 10
Explaining Diffie Hellman Key Exchange
What is Key Exchange?
Key exchange allows 2 people/parties to establish a set of common cryptographic keys without an observer being able to get these keys. Generally, to establish common symmetric keys.
How does Diffie Hellman Key Exchange work?
Alice and Bob want to talk securely. They want to establish a common key, so they can use symmetric cryptography, but they don't want to use key exchange with asymmetric cryptography. This is where DH Key Exchange comes in.
Alice and Bob both have secrets that they generate, let's call these A and B. They also have some common material that's public, let's call this C.
We need to make some assumptions. Firstly, whenever we combine secrets/material it's impossible or very very difficult to separate. Secondly, the order that they're combined in doesn't matter.
Alice and Bob will combine their secrets with the common material, and form AC and BC. They will then send these to each other, and combine that with their secrets to form two identical keys, both ABC. Now they can use this key to communicate.
Extra Resources
An excellent video if you want a visual explanation is available here. https://www.youtube.com/watch?v=NmM9HA2MQGI
DH Key Exchange is often used alongside RSA public key cryptography, to prove the identity of the person you're talking to with digital signing. This prevents someone from attacking the connection with a man-in-the-middle attack by pretending to be Bob.
Answer the questions below
I understand how Diffie Hellman Key Exchange works at a basic level
No answer needed
Correct Answer

```
root@ip-10-10-18-189:~# gpg message.gpg
gpg: WARNING: no command supplied. Trying to guess what you mean ...
gpg: encrypted with 1024-bit RSA key, ID 2A0A5FDC5081B1C5, created 2020-06-30
"TryHackMe (Example Key)"
gpg: WARNING: no command supplied. Trying to guess what you mean ...
gpg: encrypted with 1024-bit RSA key, ID 2A0A5FDC5081B1C5, created 2020-06-30
"TryHackMe (Example Key)"
```

Result:

Thus, all the required flags were successfully obtained while completing the Encryption Crypto 101 room on the TryHackMe platform.